BUILD & FLASH GUIDE

MUC Desk Radar

Everything needed to build a second one from scratch — for a friend, or a spare. Two parts, seven signal wires, a handful of free libraries. No soldering iron strictly required if you use pre-tinned header pins and Dupont jumpers. Follow it top to bottom and you'll have a glowing radar in an afternoon.

`2 components` `7 signal wires` `powered over USB-C` `Arduino IDE flash`

01 BILL OF MATERIALS

What you need

That's the whole shopping list. Both are cheap, common modules — buy a spare of each.

1 ESP32-C3 Super Mini

HW-466AB · USB-C

The brain. A tiny RISC-V microcontroller with 2.4 GHz Wi-Fi, a USB-C port for power & flashing, and an onboard button we reuse as the page control.

2 GC9A01 round display

1.28″ · 240×240 · SPI

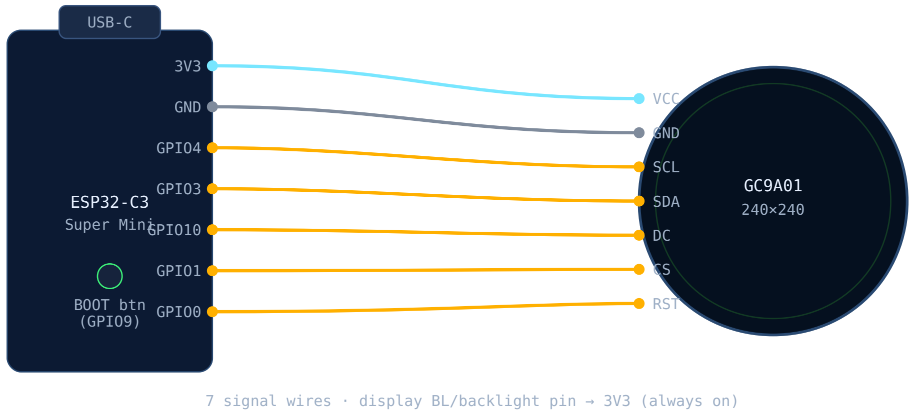
The face. A round IPS LCD driven over 4-wire SPI. Make sure it's the **GC9A01** controller version with the 7-pin SPI header (VCC, GND, SCL, SDA, RST, DC, CS).

> Plus: a USB-C cable, and a few thin jumper wires (about seven). Optional: an external momentary push-button if you'd rather not use the onboard one.

02 SCHEMATIC

Wiring it up

Seven wires between the two boards. The display runs on **3.3 V** logic and power — the same rail the Super Mini already provides. Match the table exactly.



ESP32-C3 Super Mini → GC9A01 1.28″ round display

DISPLAY PIN	WIRE TO ESP32-C3	ROLE
VCC	3V3	Power — **3.3 V only**
GND	GND	Ground
SCL	GPIO4	SPI clock (SCLK)
SDA	GPIO3	SPI data (MOSI)
DC	GPIO10	Data/command select
CS	GPIO1	Chip select
RST	GPIO0	Reset
BLK / BL	3V3	Backlight — tie high for always-on

Do not feed the display 5 V. VCC goes to the Super Mini's 3V3 pin, never to 5V/VBUS. The panel is a 3.3 V part and 5 V can kill it.

Page button. By default the firmware uses the Super Mini's onboard **BOOT** button (GPIO9) — zero extra wiring. Prefer an external button? Wire a momentary switch between GPIO2 and GND and set PAGE_BUTTON_PIN 2 in your config.

03 FIRMWARE

Flashing the firmware

Done once with the free Arduino IDE. Wi-Fi credentials live in a small `config.h` that never leaves your machine.

1 (2.x) and add ESP32 board support: in `Preferences → Additional Boards Manager URLs` add Espressif's ESP32 URL, then in install .

2 from Library Manager: `Adafruit GFX`, `Adafruit GC9A01A`, and `ArduinoJson` (v6+).

3 `firmware/plane_radar_v1/plane_radar_v1.ino`.

4 Copy `config.example.h` → `config.h` in the same folder, then fill in the three required blocks: your Wi-Fi name & password, your latitude/longitude, and your home airport. Everything else can stay on its defaults.

5 Tools → Board → `ESP32C3 Dev Module` (or "ESP32-C3 Super Mini" if listed). Set `USB CDC On Boot: Enabled` so the serial monitor works over USB-C.

6 the Super Mini over USB-C and pick its port under Tools → Port.

7 Hit the arrow. If it won't start, hold `BOOT`, tap `RST`, release BOOT to force the bootloader, then upload again.

8 You'll see the radar ping animation, your name, a Wi-Fi line, then live traffic within a few seconds. Done.

> **Serial monitor** at 115200 baud is your friend: it prints the assigned IP, fetch status, and any HTTP/JSON errors while you get the wiring dialled in.

04 WHEN IT MISBEHAVES

Troubleshooting

SYMPTOM	LIKELY CAUSE & FIX
Blank / white screen	Re-check the 7 wires and confirm VCC is on 3V3. Make sure the backlight (BL) pin is tied to 3V3. Swap SDA/SCL if mixed up.
Screen on but garbled / wrong colours	DC or RST on the wrong GPIO, or a loose jumper. Verify GPIO10=DC, GPIO0=RST, GPIO1=CS. Some panels need a colour-invert tweak in the display init.
Upload fails / port vanishes	Force the bootloader: hold `BOOT`, tap `RST`, release BOOT, upload. Try another USB-C cable — some are power-only.
Stuck on "WIFI FAILED"	The chip is 2.4 GHz only — a 5 GHz-only SSID won't connect. Check the name/password capitalisation in `config.h`. The device auto-reboots and retries by design.
"JSON fail" / "HTTP −11" pill	A hiccup on the free feed or Wi-Fi. It retries automatically after a few seconds and keeps showing the last good traffic.
"API busy" pill	Rate-limited (HTTP 429). The firmware backs off ~45 s on purpose and resumes. If frequent, raise `FETCH_INTERVAL_MS`.
Button does nothing	If using the onboard button, confirm `PAGE_BUTTON_PIN 9`. For an external one, it wires to `GPIO2` and `GND` with `PAGE_BUTTON_PIN 2`.
Chip enters flashing mode at power-up	You held the onboard BOOT button while plugging in USB. Release it and re-plug.
	Harmless feed quirk (non-ASCII bytes) — the firmware already filters these; a reboot clears any stale card.

Text shows
blocks /
@@@@

05 MAKE ONE FOR A FRIEND

Handing it on

To gift a built unit, the recipient only has to make it theirs — no rebuild needed:

1. They copy `config.example.h` → `config.h` and set their and . Two required edits, really.

2. Optionally set `OWNER_NAME` for the boot splash and their if it isn't Munich.

3. Re-flash once. Everything else — pages, colours, timings, scoring — has sensible defaults baked in.

> Because `config.h` is the only file with personal details and it's kept out of version control, you can share the whole project freely without leaking anyone's Wi-Fi password or address.